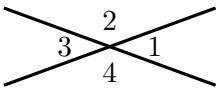
Proving Angle Relationships — Practice

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Vertical Angles and Linear Pairs

Watch Me — Worked Example



$m\angle 1 = 48°$. $\angle 3$ is vertical to $\angle 1$: $m\angle 3 = 48°$.
$\angle 2$ and $\angle 1$ are a linear pair: $m\angle 2 = 180° - 48° = 132°$; $m\angle 4 = 132°$ (vertical to $\angle 2$).

1. Use the figure above. $m\angle 2 = 115°$. Find $m\angle 4$.

2. $m\angle 2 = 115°$. Find $m\angle 1$.

3. $m\angle 3 = 72°$. Find $m\angle 1$, $m\angle 2$ and $m\angle 4$.

4. Are $\angle 1$ and $\angle 2$ vertical angles or a linear pair?

Type 2 Solve With Algebra

Watch Me — Worked Example

$m\angle 1 = 4x + 6$ and $m\angle 3 = 6x - 14$ (vertical angles). $4x + 6 = 6x - 14 \Rightarrow 20 = 2x \Rightarrow x = 10$.
$m\angle 1 = 4(10) + 6 = 46°$. **Answer: $x = 10$, $m\angle 1 = 46°$**

5. $m\angle 1 = 5x - 12$, $m\angle 3 = 3x + 20$. Find x and $m\angle 1$.

6. $m\angle 1 = 2x + 10$, $m\angle 2 = 4x - 10$ (linear pair). Find x and $m\angle 2$.

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7. $\angle A$ and $\angle B$ are both complementary to $\angle C$. $m\angle A = 3x - 4$, $m\angle B = x + 20$. Find $m\angle A$.

8. HARD $m\angle 1 = x^2$, $m\angle 3 = 5x + 14$ (vertical angles), $x > 0$. Find $m\angle 1$.

Type 3 Complete the Proof

Watch Me — Worked Example

Given: $\angle 1$ and $\angle 2$ are right angles. Prove: $\angle 1 \cong \angle 2$.
1. $\angle 1$ and $\angle 2$ are right angles (Given) 2. $\angle 1 \cong \angle 2$ (Right Angles Congruence Theorem)

9. Given: two lines intersect, forming $\angle 1$–$\angle 4$ as in the Type 1 figure.
Prove: $\angle 2 \cong \angle 4$.

Statements	Reasons
1. $\angle 2$ and $\angle 4$ are vertical angles	
2. $\angle 2 \cong \angle 4$	

10. Given: $\angle 1$ and $\angle 2$ are a linear pair, and $\angle 2$ and $\angle 3$ are a linear pair. Prove: $\angle 1 \cong \angle 3$.

Statements	Reasons
1. $\angle 1, \angle 2$ and $\angle 2, \angle 3$ are linear pairs	Given
2. $\angle 1, \angle 2$ supplementary; $\angle 2, \angle 3$ supplementary	
3. $\angle 1 \cong \angle 3$	

Type 4 Always, Sometimes or Never?

Watch Me — Worked Example

"Two angles that form a linear pair are congruent." Only when both are $90°$. **Answer: sometimes.**

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11. Vertical angles are congruent.

12. Two right angles are supplementary.

13. Two supplementary angles are both acute.

14. HARD Two angles complementary to the same angle are congruent.

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